

I. INTRODUCTION

This is for...

II. EXTENDED LITERATURE REVIEW

A. Complexity of Clinical Narratives

Clinical letters use narrative language so that clinicians can record the patient's situation, their judgement, and their uncertainty. That flexibility supports clinical communication, but it makes the information difficult to use at scale [1]. Clinicians often use hedging to describe certainty, frequency, and quantity rather than stating a precise value [2]. The task of information extraction (IE) is to convert that nuanced information into a standardised form while preserving its essential meaning.

Epilepsy clinic letters illustrate that complexity. One letter can contain diagnoses, prescriptions, history, seizure descriptions, and several references to seizure frequency. Fonferko-Shadrach *et al.* report much lower human agreement for seizure frequency (per-item F1 = 0.47) than for prescriptions (0.87) or diagnoses (0.83) [3]. They attribute this to the variety of frequency expressions and to mentions of more than one seizure or event type. Effective extraction requires clearly defined policies and procedures describing what to extract and how to represent it.

B. Seizure Frequency as a Selection Task

The same letter can be used for different purposes. ExECT and the later annotation study treat it as an inventory: they record several entity mentions and their attributes. One letter may list several anti-seizure medications, each with a name, dose, unit, and administration frequency [3], [4]. That form is appropriate when the aim is building a broad clinical inventory.

The task in this study has a different purpose. Gan *et al.* assign each letter one label for the patient's current seizure frequency status [5]. The system has to gather candidate descriptions, interpret their meaning and time frame, and choose the single normalised label that best represents the current pattern. Earlier one-label studies use the same formulation [6]–[8]. The single label extraction implies that it is all one task, but this paper proposes separate stages for finding candidate evidence, encoding it, and selecting the final answer.

Two common cases show why that split matters. Diary-style counts may need to be combined: *1 in June, 0 in July, 2 in August* can become *3 in 3 months*. Competing statements may instead require precedence: *She may remain seizure-free for up to 4 months, but then will experience clusters of 5 seizures in a single day*. Different annotation guidelines would record those cases differently. Separate stages make the policy choice inspectable rather than implicit.

C. Rules, Fine-Tuned Models, and LLMs

Several papers have published seizure frequency extraction systems using rules, fine-tuned models, or prompted large language models. Rules match terms and apply domain logic. ExECT does this for a clinical inventory; Decker *et al.* use regular expressions for seizure-related variables [4], [9]. Fine-tuned BERT models learn the mapping from labelled

examples, as in Xie *et al.* and Fernandes *et al.* [6], [10]. Table III lists the F1 scores those studies report. The scores are not a ranking of methods: each study uses its own dataset, label set, and evaluation.

Prompted and fine-tuned LLMs are more recent. Holgate *et al.* show that a frozen model can extract seizure information from instructions and examples [7]. Later work, including Holgate *et al.* and Gan *et al.*, fine-tunes a model to a defined output format [5], [8].

BERT-style encoders and decoder-based LLMs share the transformer architecture. The practical difference is in-context learning: a LLM can follow instructions and examples without a task-specific training run [11]. That behaviour transfers to clinical extraction, where clear instructions and examples can reduce the need for heavy post-processing [12]. Gan *et al.* show that combining structured prompting and fine-tuning can deliver even stronger performance. This paper explores whether structured prompting in a modular system can deliver comparable performance to a fine-tuned LLM.

D. Modular and Sequential Extraction Pipelines

Multi-stage clinical extraction pipelines split a task into distinct operations instead of asking a model for one final answer in a single step. CLINES uses stages for chunking, extraction, attribute assignment, normalisation, date resolution, and aggregation [14]. Seizure frequency studies have also decomposed the task in structured prompting including numbered reasoning steps and summarise-then-extract designs [7], [15].

This study decomposes the task into three stages: finding candidate evidence, encoding it into a structured form, and selecting the final answer. Each stage is part of a modular pipeline and each module can use rules, an LLM call, or a hybrid of both. Past research on seizure frequency uses either LLMs or rules at one stage. This paper empirically evaluates each of them plus a hybrid combination of the two across all three stages.

III. PLANNING

The project lifecycle is divided into five stages: planning, development, implementation, testing and evaluation. The sections below describe the key decisions, the rationale behind them, and the sources they were based on, in chronological order as far as possible.

There are two key documents in the planning phase:

- 1) The project brief, created by my supervisor.
- 2) The problem specification and project plan, created by me.

A. Project Brief

The project brief outlines the initial scope of the project. The problem specification and project plan refines the scope, specifies the research objectives and describes the methods used to achieve them. The project brief was:

Instead of training models, this project builds a training-free (or minimal-training) multi-agent extraction system that reads epilepsy letters and outputs structured fields (e.g., ASM, seizure type/frequency, investigations, epilepsy syndrome/type). The student will design roles such as: (a) Section/Time-line Agent (segments text, builds timeline), (b) Field Extractor Agents (one per field group), (c) Verification Agent (checks evidence spans, contradictions, missingness), and (d) Aggregator Agent (produces final JSON + confidence + citations to text spans). The key research goal is reliability: compare single-prompt extraction vs multi-agent extraction under the same budget constraints, and test improvements from self-consistency, evidence requirements (“answer only if supported by quote”), and structured output validation. Evaluation uses field-level accuracy/F1, and robustness tests. Data can be synthetic or de-identified samples if available. Deliverables: an end-to-end agentic pipeline, evaluation harness, and a dissertation focused on engineering + empirical study of reliability.

The deliverables are a narrow specification of what a successful project would look like:

- 1) An end-to-end agentic pipeline.
- 2) An evaluation harness.
- 3) A dissertation focused on engineering + empirical study of reliability.

The type of data this would be applied to was narrowly defined as epilepsy letters, but it was undetermined whether the specific data source was synthetic or de-identified samples. This was one key decision to be agreed in the project kick-off meeting.

The research would compare a single-prompt extraction vs. multi-agent extraction, and the multi-agent system included a number of initial suggestions such as verification agent and aggregator agent. The precise implementation of the multi agent system would be defined as the research progresses, guided by best practices in the literature and the results from empirical experiments.

In the initial project kick-off meeting it was agreed that the specific data source used would be a set of 1,500 synthetic clinical letters that the supervisor had access to. As these were synthetic letters, there would be no need to get specific ethics approval to use for research purposes.

B. Problem Specification and Project Plan

The problem specification and project plan was uploaded to the university Canvas site on the 5th June 2026. This is a direct extract from the methods section:

There are two primary data sources in this project:

- 1,500 synthetic clinical letters and structured labels from Gan *et. al* [5]
- 200 synthetic clinical letters and structured labels from Fonferko-Shadrach *et. al* [3]

The project will use LLMs as the core AI tool, using both open (e.g. Qwen 3.6) and closed (e.g. GPT 5.4) models. The research will involve building an agent harness around the model that enables it to ingest clinical letters and prompt instructions, produce a structured output, and have that output validated and evaluated.

The research will be empirical in nature and focus primarily on assessing the quality, reliability and validity of the outputs. Cost, latency, data schema complexity and evidence specificity will be key experimental variables tested.

Evaluation will take place in three key forms: comparing the accuracy (F1 score) of the outputs vs. the ground-truth labels from the two data sources, assessing the validity of the structured data schema created, and assessing the validity of the evidence cited. A series of ablations will be conducted to isolate the impact of different components of the agent harness.

The F1 score evaluated on a set of ground-truth labels is the de-facto standard for evaluating AI models on how well they can complete a classification task. Evaluating the model on two different datasets optimised for two different tasks effectively tests the flexibility of the system.

The ExECTv2 dataset [3] was used extensively as per the original plan. The pipeline was built, experiments were conducted and results were analysed in depth and benchmarked against the prior results. However, when the time came to write the paper it became clear that writing about two datasets with different annotation policies, evaluation measures, error profiles and sample sizes required the reader to hold too much information in mind.

As a result, the empirical analysis in the paper and in these materials focus exclusively on the seizure frequency extraction task. However, we used the pipeline built for ExECTv2 to extract diagnoses, medications and investigations from the main dataset of 1,500 synthetic letters as a feasibility study. Examples of that are shown in the appendices here, and some of the design choices for the seizure frequency extraction pipeline are directly informed by designing a modular system that can be applied to both tasks. Results benchmarked against the ExECTv2 dataset will be published in a separate paper.

The following sections describe how the rest of this plan was implemented.

IV. DEVELOPMENT

This section explains how the initial project proposal was translated into a focused and testable research approach. It records the evidence, design choices, and trade-offs that led to the final method.

A. Original Design Requirements

The original brief contained a series of key design requirements for the pipeline. It must:

- 1) Ingest epilepsy letters
- 2) Output validated structured fields
- 3) Contain verified evidence citations
- 4) Produce this in an aggregated JSON

These design requirements were tightly coupled to a multi-agent framework, e.g using a *Verification Agent* to check evidence spans and an *Aggregator Agent* to produce the final JSON and citations.

The literature review showed that multi-agent systems are particularly useful for complex tasks [25], especially on long-context tasks [26]. However, research also shows that single-agent systems are often as effective when they are given equal compute and multi-agent systems create new failure modes as errors propagate throughout the system [26].

For this task, the full epilepsy letter and instructions fit comfortably in the LLM’s context window so there is no need for convoluted multi-hop pipelines or document splitting that multi-agent systems performed particularly well on [14]. Thus, a multi-agent system can not be justified on that basis.

It is less clear whether this task is sufficiently complex to justify a multi-agent system. Past research has shown that LLMs struggle with temporal logic and numerical logic, and both occur in this task. It is also a task that human annotators struggle to do reliably and consistently [3]. This suggests the task is at least somewhat complex and a multi-agent system is justified on that basis.

However, it is important that the each agent is given a sufficiently complex task to justify the added system complexity. A key goal of the research is to design a reliable system. A reliable system is one that contains the minimum amount of complexity necessary to function. More components means creates more failure modes, and in a sequential process those errors propagate and multiply at each step.

B. Deterministic Code for Deterministic Tasks

A key design choice early on was to simplify. Any task that can be reliably completed by a deterministic system (code) instead of a probabilistic system (LLMs) should be done deterministically.

Evidence verification and schema validation are two tasks that code can do reliably and efficiently. Initial experiments showed that LLMs can not do this task reliably; they introduce as many new schema errors as they repair, and they occasionally miss invalid evidence. However even if they could do the task reliably, code already does this at a much lower cost and latency.

Therefore a dedicated *Aggregator Agent* to produce the final JSON with text citations and a dedicated *Verification Agent* to check evidence spans were removed from the design after initial experiments. A *Field Extractor* agent was no longer relevant as the task was limited to a single field, seizure frequency. Initial experiments showed that a *Section/Timeline Agent* was a net-negative contribution because correct seizure frequency references were not always in the dedicated seizure frequency section, and splitting up the text encouraged the LLM to select wrong answers more frequently.

C. Pipeline Design Review

After initial experiments, all four of the original proposed LLM agents had been ruled out because they were now either invalid or empirically performed worse than a single LLM. This led to a major review of the pipeline design.

Comprehensive error analysis showed that the LLM was consistently finding valid seizure frequency patterns, providing exact evidence from the text and often choosing defensible answers based on the instructions. It did not another agent to check its evidence or structure it into JSON, it needed cleaner task separation.

The task of extracting the correct seizure frequency label is composed of three sub-tasks: finding relevant mentions of seizure frequency in the text, encoding that information into the form the gold-standard labels expect, and applying the a policy to choose which of those labels aligns matches the gold-standard selected label. The labelling conventions and policy choices in the gold-standard evidence are valid, but they are not the only valid choice. Therefore a key pipeline requirement is producing results that conform to an explicit set of subjective choices.

The original multi-agent pipeline separated mechanical tasks across multiple LLMs; one agent splits the evidence, another checks the evidence, another aggregates it together. With simplicity and efficiency in mind, it would be preferable if a single LLM could do all of those tasks at once. This would require the LLM to generate a much richer intermediate schema. Rather than extracting a single seizure frequency label, the LLM would extract every relevant seizure frequency pattern as a distinct entity in a nested JSON, with exact evidence for every entity, and make an initial proposal.

Using this richer schema, a subsequent LLM could then review the evidence and apply the correct policy in hard cases. This would reframe the multi-agent pipeline into separate agents for separate clinical tasks instead of mechanical tasks. This richer schema and three stage pipeline also makes it possible for deterministic rules to apply policies to the evidence generated by the LLM to select the final answer.

Past research shows that deterministic code can reliably normalise seizure frequency values, and hybrid LLM and rules systems can outperform LLMs alone. Rules-based systems struggled with seizure frequency extraction more than medications, but they still performed better than human experts. Deterministic code already performed schema validation and evidence verification more reliably than LLMs, so maybe there was a role for more deterministic code in the pipeline.

D. Final Pipeline Design

The final outcome of this design review was a modular pipeline that could use either LLMs or rules at each of the three stages *find*, *encode* and *select*. These are the implementation names; the paper reports *find* and *encode* together as *extract* and calls *select* *decide*. Three methods would be implemented: a multi-agent pipeline, a purely rules-based system and a hybrid LLM and rules system.

The original project brief required a pipeline that could ingest epilepsy letters, output validated structured fields, contain verified evidence citations and produce this in an aggregated JSON. It required an evaluation harness and a dissertation focused on engineering and an empirical study of reliability. This design meets all of these requirements.

The 3-method comparison strengthens the empirical core and deepens the focus on reliability by using the evaluation harness to determine whether a multi-agent pipeline is the most reliable pipeline.

V. IMPLEMENTATION

The implementation was designed to produce a structured, reproducible account of current seizure frequency from a clinical letter, while retaining the evidence and intermediate choices needed to inspect how an answer was reached. It describes the reusable pipeline and the controlled comparison design; performance results and the findings of individual ablations are reported in the evaluation section.

A. Structured Record

Each method receives a synthetic clinical letter and produces a structured seizure frequency record. The final record contains the submitted frequency label, direct supporting evidence, and the metadata needed to identify the method, model configuration, data split, and scoring policy used. This design makes the final answer usable for benchmark evaluation without reducing the whole process to an opaque single-label extraction.

The pipeline uses a richer intermediate record than the final benchmark label. During extraction, it can retain multiple candidate seizure frequency events, their source spans, raw wording, time information, and an initial proposed answer. This is important because a letter may contain several plausible statements: historical and current rates, periods without seizures, isolated breakthrough events, clusters, and uncertain or qualitative descriptions. Preserving the alternatives avoids losing information before the annotation policy has been applied.

B. Find, Encode and Select

The record distinguishes the source wording from its encoded representation and from the final selected answer. A later selection stage can revise the proposed answer while retaining the evidence on which the original proposal was based. The final output is validated against the expected schema, while source data and raw model output are retained separately from format repair and clinical policy decisions.

The implemented pipeline divides seizure frequency extraction into three stages. *Find* collects candidate events and their evidence from the letter. *Encode* represents an already identified candidate in the required structured form, including the benchmark’s label conventions. *Select* applies the policy for determining which candidate best represents the patient’s current seizure frequency pattern. This division makes explicit that locating a relevant statement, writing it in the desired

form, and choosing the correct final statement are different tasks.

Each stage produces an inspectable output. This is particularly useful when the selected answer changes: the change can be attributed to the evidence found, its encoding, or the final policy decision. It also permits later stages to work from the same saved candidate record, rather than requiring every alternative implementation to make a new model call. This significantly sped up the optimisation process.

C. Rules and Deterministic Components

Deterministic components implement tasks for which explicit, reproducible handling is appropriate. These include parsing and schema repair, validation of required evidence fields, aggregation of the final record, and rules for normalising frequency expressions into the task’s closed vocabulary. The implementation keeps format-only repair, benchmark representation, and clinically meaningful policy rules distinct so that a change to one does not silently alter another.

Rules also provide an implemented option at each substantive stage. They can identify seizure-frequency statements, encode candidate events, and apply the final selection policy. Their authority is recorded by rule type: for example, whether a rule merely standardises a representation, blocks an unsupported candidate, rewrites a submitted concept, or reselects between already found events. This makes deterministic processing an explicit part of the method rather than unreported post-processing.

D. LLM Components

The LLM implementation uses structured prompts that ask the model to return a record containing candidate seizure-frequency events, evidence, and a proposed answer. LLMs can be assigned to *find*, *encode*, *select*. At *find*, an LLM identifies candidate events and evidence. At *encode*, it maps an already represented candidate to the required label form. At *select*, it adjudicates between competing candidates under the same policy that deterministic rules may apply.

This stage-assignment approach provides a controlled way to compare rules and LLMs on the substantive clinical extraction decisions, while deterministic validation and output assembly remain common infrastructure.

VI. TESTING

Testing verified that the implemented system and research controls behaved as intended before its comparative results were interpreted.

Data-loading tests verify that source records, gold labels, evidence references, quality flags, and label normalisation are preserved. The test suite also checks that the fixed split manifest contains the intended disjoint training, development, and held-out partitions, and that loaders return records in the manifest-defined order. These checks protect against accidental changes to the dataset surface, label interpretation, or the membership of a reported split.

The split policy was treated as a testable research safeguard rather than a documentation convention. Development work

used the permitted development partition for prompt and rule changes, while the held-out partition could be used only under a fixed protocol and reported at aggregate level. Tests enforce that held-out replay artifacts do not retain row identifiers, source notes, labels, predictions, or error lists.

Focused regression tests capture failure modes identified during development. These include aggregating diary entries across months, avoiding double-counting when the same evidence is repeated or split across candidates, handling zero-count months, interpreting clusters, and distinguishing current seizure burden from historical or less relevant statements. Each fixture represents a named clinical or annotation-policy obligation, rather than a generic collection of examples. These tests were used whenever a prompt, rule, parser, or stage boundary changed. Their purpose was to show that a local improvement did not silently regress previously correct behaviour.

The implemented methods were tested as pipelines as well as individual functions. Prompt-contract tests check that model requests preserve the required structured fields and do not inadvertently introduce irrelevant metadata. Pipeline and runner tests check that the intended components run in the intended order and that their outputs are written into the common record structure.

Together, these tests verify that the implementation produces valid records, applies the intended data and split controls, and preserves known clinical-policy behaviour.

VII. EVALUATION

This section evaluates whether the verified pipeline met the research objective: to extract a patient’s current seizure frequency burden from a clinical letter reliably, with a structured evidence record and a reproducible comparison between rules and LLMs.

A. Finding Multiple Candidates with Exact Evidence

A key feature of the pipeline was the rich schema. Models were asked to extract multiple seizure frequency patterns in the text, extract evidence spans for each pattern identified, and choose which of those patterns best represents the patient’s current seizure frequency status. That rich schema is what enabled a subsequent LLM in the multi-agent method and rules in the hybrid method to re-use the evidence and deliver performance gains of +0.07. However that comes at the cost of higher schema complexity and task specificity and this is where local models remain slightly worse.

Gemini, Grok, Luna and DeepSeek all had exact evidence rates of 99%+. Qwen (97%) and Gemma (93%) occasionally paraphrased the text. Gemini, Grok and Luna produced valid schema 100% of the time. DeepSeek and Qwen produced valid schema 99% of the time. Gemma was only at 97%, occasionally using things like double `{{` or `'''` and breaking the nested json structure.

B. Multi-Agent Pipeline Variants

C. LLM Prompt Variants

D. Hybrid Pipeline Variants

Rules and LLM finds,

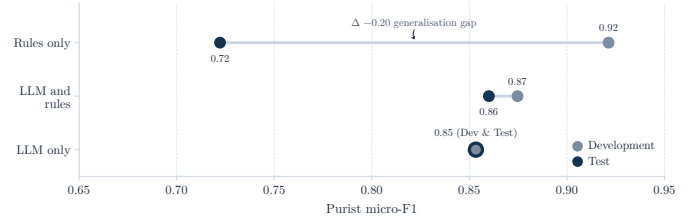


Fig. 1: Rules only performs best on the development set but overfits. LLM only is 0.85 on both splits under the living select prompt. LLM and rules remain the strongest holdout row, with a generalisation gap of -0.01 (0.87→0.86).

VIII. DEVELOPMENT VS. TEST RESULTS

Rules only perform better on the development set but significantly worse on the test set (Figure 1), which repeats the pattern of overfitting commonly found in the literature. Rules are brittle and tend to overfit specific patterns in the training data, generalising poorly to unseen data. The Rules only method achieves an F1 score of 0.92 on the development set, but performance declines significantly (-0.20) on the test set. LLM only is 0.85 on both splits which shows it generalises particularly well. The LLM and Rules method combines the best parts of each: it performs better on the development set than LLM only (+0.02) and it has a significantly smaller generalisation gap than rules only (-0.01). The LLM does the heavy lifting of matching patterns across the entire text, allowing the rules to focus on generalisable patterns in the refined evidence set.

In the early phases of development, comprehensive error analysis showed that the LLM was consistently choosing valid answers based on the instructions. For example, the LLM is told to choose the seizure frequency with the highest current or recent seizure pattern, if multiple seizure events are available. When it sees a pattern such as *2 in Jan, 0 in Feb, 0 in March, 5 in April* the model chooses 5 because it is the largest and most recent pattern. However the gold-standard label is 7 in 4 months. When the prompt instruction was changed to add up all seizure events in a period, then it combined tonic-clonic seizures and generalised seizures over overlapping periods, while the gold-standard choose the highest count among the two seizure types. Separating these complex, somewhat contradictory instructions into a separate LLM prompt or rules at the select phase produced significant improvements (+0.06).

IX. TOOLS DESCRIPTIONS

X. CRITICAL APPRAISAL

A. Successes

B. Challenges

The method is available as an open source package here along with a web app demonstrating key features of the pipeline here .

Daniel Carter, DOB: 17-03-1982, Hospital No: KCH902311 NHS No. 9483726150
12 Brookfield Road, London, SE15 2HQ

Thank you for asking me to review your patient, whom I saw in clinic today with his partner. He described a long history of intermittent episodes that have varied over the years. The current concerns relate to brief spells of behavioural arrest with a sense of pressure in the chest and a tendency to pace or fidget, usually lasting a few minutes, without clear loss of awareness, and without tongue biting or incontinence. He can feel unsettled afterwards but returns to baseline within 10-15 minutes. He has not reported injuries or nocturnal events in recent months.

He had previously been referred after a period of frequent episodes during a stressful time at work. Since engaging with psychological support and using relaxation techniques advised by our liaison psychology team, his day-to-day anxiety has eased. In that context, events at present are considered non-epileptic and have become more manageable now that his anxiety is better controlled. He and his partner keep a simple diary which shows fewer episodes over the past six weeks, and they both feel more confident about identifying triggers and using breathing strategies early.

There is no clear history to suggest a specific seizure phenotype, and prior routine EEG and MRI from last year were unremarkable. He remains on a stable dose of Brivaracetam initiated previously by another service; he has not had adverse effects and prefers no change at this stage. We discussed the potential value of further characterisation should the picture alter, including ambulatory EEG or admission for video-EEG telemetry if clinically indicated, but for now he is content to continue with self-management and follow-up.

```
select selected: e1
no seizure frequency reference
"There is no clear history to suggest a specific seizure phenotype"

selected_ids: ["e1"]
rationale: "The note discusses non-epileptic episodes and notes no clear history of epileptic seizures, with no usable seizure frequency reference."
```

Fig. 2: Seizure free → unknown error example. Gold-standard label: seizure free for multiple month. Model selected no seizure frequency reference (unknown).

According to the colleague, the typical event begins without warning while he is plating or working over the grill. He becomes unresponsive, stares ahead, and his posture slackens; there may be low, indistinct vocal sounds. On two occasions he slid down a stainless-steel unit and briefly rocked at the shoulders. There has been no clearly observed rhythmic limb jerking, tongue biting, or incontinence reported. He appears confused after the episode and needs 30-60 minutes before he can resume tasks, often describing a heavy head and overwhelming fatigue. He does not reliably recall the event itself.

Daniel Harris reports that these episodes crop up most shifts, especially during the busiest part of service, and his manager has had to reassign him from open flame stations as a precaution. He also notes that, on getting home, he sometimes wakes disoriented on the sofa with no memory of lying down, suggesting possible unwitnessed events outside work as well. Heat, noise, and pressure to keep pace may be contributory triggers from his perspective.

```
select selected: e1
unknown
"these episodes crop up most shifts"

selected_ids: ["e1"]
rationale: "Episodes/events are described as occurring 'most shifts' / 'on most working days', which indicates active events but without a precise countable frequency."
```

Fig. 3: Frequent → unknown error example. Gold-standard label: multiple per week. Model selected unknown.

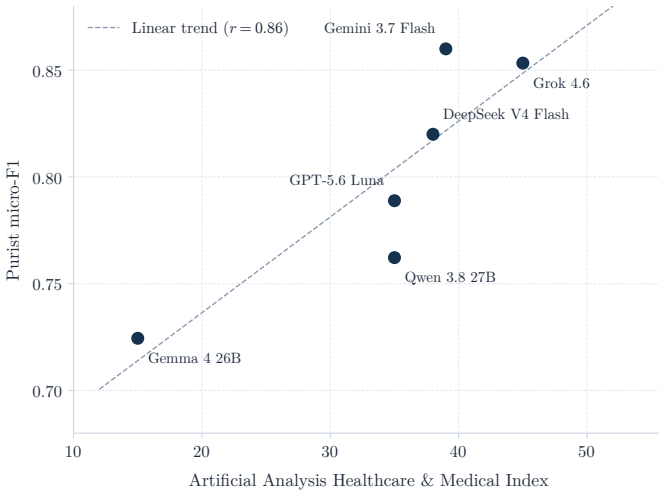


Fig. 4: The smartest models are the best frequency extractors.

TABLE I: Seizure frequency label forms

Family	Written forms
Rate	N per unit; N per N unit; N per N to N unit; N to N per unit; N to N per N unit; multiple per unit; multiple per N unit; N per multiple unit
Cluster	cluster per unit, N per cluster; cluster per unit, range per cluster; cluster per unit, multiple per cluster
Seizure free	seizure free for a duration
Unknown	unknown; no seizure frequency reference

XI. GAN CODEBOOK FIND PROMPT

The cited language-model find request is the codebook extract prompt (`gan_llm_extract`). The object below is the frozen template sent to the model: task, instructions,

TABLE II: Later-stage select examples

Events	First pick	After select
3 in March; 6 in May Last seizure on 12th March; well 3 weeks	6 per month seizure free for 3 week	9 per 3 month 1 per month

TABLE III: Comparisons to past seizure frequency papers

Author	Year	Method	F1 score
Xie et al.	2022	Fine-tuned BERT	0.83*
Decker et al.	2022	Rules	0.82**
Fonferko-Shadrach et al.	2024	Rules	0.66
Fernandes et al.	2024	Fine-tuned BERT	0.37
Holgate et al.	2024	Prompted LLM	0.73
Holgate et al.	2025	Fine-tuned LLM	0.68
Gan et al.	2026	Fine-tuned LLM	0.81
Brown & Gan	2026	LLM and rules	0.86

allowed label forms, event schema, and selection schema. The letter is attached at run time as `note_text` and is omitted here. Research metadata such as `prompt_version` is not part of the model-facing payload.

This template is not the source-near find ablation (`gan_llm_extract_raw`), which has no closed `label_forms` block. It is not the later select prompt.

The only preprocessing applied to the answer labels was to use the CUIPhrase as the main answer key for diagnosis and seizure frequency instead of the raw text value. Often they were the same label, but the CUIPhrase sometimes included cleaner labels for matching answers e.g. the normalised CUIPhrase 'temporal-lobe-epilepsy' was used over the text 'epilepsy-Probable-temporal' for the same span.

Clinical Summary:

Since our last review, the patient describes meaningful improvement in day-to-day stability. Night-time childcare support has been introduced, sleep has become more consolidated, and both the patient and partner report steadier mornings with fewer interruptions. In this context, No further tonic-clonic seizures have occurred since 06/2017, although **three single jerks remain. These jerks were brief, without loss of awareness, and occurred during periods of sleep restriction earlier in the year.** No injuries, tongue biting, incontinence, or prolonged post-event confusion have been reported. The patient continues working and notes improved concentration since sleep regularised.

select selected: e2

unknown

"three single jerks remain. These jerks were brief, without loss of awareness, and occurred during periods of sleep restriction earlier in the year"

selected_ids: ["e2"]

rationale: "The patient is not completely seizure-free because three single jerks occurred earlier in the year, but the exact frequency is vague/not standardly countable."

Fig. 5: Infrequent → unknown error example. Gold-standard label: 3 per 14 month. Model selected unknown.

XII. RESULTS

A. Development Environment

B. Methods

C. Models

D. Structured Evidence

E. Ablations

F. Error Analysis

XIII. DISCUSSION

XIV. CONCLUSION

REFERENCES

```
{
  "task": "Read the clinical note. Extract seizure-frequency facts as slim events, then select the current burden .",
  "instructions": [
    "Read the full clinical note and extract source-near seizure-frequency facts.",
    "Return events as slim clinical facts, not fully normalized answer records. Use raw_value for the text's stated rate, duration, last-event statement, or unknown/no-reference cue.",
    "Event kind must be one of frequency_rate, cluster_frequency, seizure_free, last_event_only, unknown_frequency, or no_reference.",
    "Use one no_reference event only when the note contains no usable seizure-frequency evidence. Do not use no_reference when seizures are discussed but frequency is unclear; use unknown_frequency instead.",
    "Keep seizure-free statements separate from unknown or last-event-only statements. Do not select seizure-free if other current seizure-like events remain active.",
    "Selection must choose the highest current or recent seizure burden across semilogies when several current seizure types are present.",
    "If the note gives an overall current seizure count plus a breakdown by seizure type, select the overall count for the seizure-frequency label rather than only the clinically most severe subtype count.",
    "Write the seizure-frequency label using only the allowed forms. Copy an example and change the numbers if needed.",
    "If the selected event has a countable stated rate, put the note's wording in raw_value and write the seizure-frequency label from the allowed forms.",
    "When the note says a last event occurred on a date and the patient has been well, stable, or seizure-free since, still extract the dated last-event fact as its own event even if the selection is seizure-free.",
    "When the note says a count such as 3 or 4 jerks occurred since a dated last tonic-clonic seizure, keep the source count and the dated anchor available in the event list.",
    "Every evidence value must be an exact substring from the note when possible.",
    "Return exactly one JSON object with no markdown."
  ],
  "label_forms": {
    "rules": [
      "Write the label using only the forms below. Copy an example and change the numbers if needed.",
      "Use digits, not word numbers.",

```

```

      "Use day, week, month, or year. Prefer the singular word.",
      "A night count is a day count: once per night becomes 1 per day.",
      "Flatten a bound: at most four per day, or ≤ four per day, becomes 4 per day.",
      "Do not write <=, >=, up to, at most, several, handful, frequent, or a calendar date.",
      "If none of these forms fit, write unknown when seizures are discussed, or no seizure frequency reference when there is no usable frequency evidence."
    ],
    "forms": [
      {
        "form": "N per unit",
        "description": "A single count in one day, week, month, or year.",
        "examples": [
          "1 per day",
          "4 per week",
          "3 per month",
          "2 per year"
        ]
      },
      {
        "form": "N per N unit",
        "description": "A single count over a stated number of days, weeks, months, or years.",
        "examples": [
          "1 per 2 day",
          "1 per 2 week",
          "2 per 6 month"
        ]
      },
      {
        "form": "N per N to N unit",
        "description": "A single count over a range of days, weeks, months, or years.",
        "examples": [
          "1 per 2 to 3 day",
          "1 per 2 to 3 week",
          "1 per 4 to 6 month"
        ]
      },
      {
        "form": "N to N per unit",
        "description": "A count range in one day, week, month, or year.",
        "examples": [
          "2 to 3 per day",
          "2 to 4 per week",
          "3 to 5 per month",
          "2 to 4 per year"
        ]
      },
      {
        "form": "N to N per N unit",
        "description": "A count range over a stated number of days, weeks, months, or years.",
        "examples": [
          "2 to 4 per 3 month",
          "2 to 3 per 2 week"
        ]
      },
      {
        "form": "multiple per unit",
        "description": "More than one seizure in one day, week, or month, with no number given.",
        "examples": [
          "multiple per day",
          "multiple per week",

```

```

    "multiple per month"
  ],
  {
    "form": "multiple per N unit",
    "description": "More than one seizure over a stated
number of months, with no number given.",
    "examples": [
      "multiple per 2 month"
    ]
  },
  {
    "form": "N per multiple unit",
    "description": "A count over an unstated number of
days or months.",
    "examples": [
      "1 per multiple day",
      "1 per multiple month"
    ]
  },
  {
    "form": "cluster per unit, N per cluster",
    "description": "How often clusters happen, and how
many seizures are in each cluster. The cluster count or
the time between clusters may be a range.",
    "examples": [
      "1 cluster per day, 5 per cluster",
      "1 cluster per week, 4 per cluster",
      "1 cluster per 4 month, 5 per cluster",
      "6 cluster per month, 4 per cluster",
      "1 cluster per 4 to 5 day, 2 per cluster"
    ]
  },
  {
    "form": "cluster per unit, range per cluster",
    "description": "How often clusters happen, and a
range of seizures in each cluster.",
    "examples": [
      "1 cluster per day, 2 to 4 per cluster",
      "1 cluster per week, 3 to 6 per cluster",
      "1 cluster per month, 4 to 6 per cluster"
    ]
  },
  {
    "form": "cluster per unit, multiple per cluster",
    "description": "How often clusters happen, and more
than one seizure in each cluster with no number given
.",
    "examples": [
      "1 cluster per day, multiple per cluster",
      "1 cluster per week, multiple per cluster",
      "1 cluster per month, multiple per cluster"
    ]
  },
  {
    "form": "unknown cluster count",
    "description": "Clusters are described, but how
often they happen is not known.",
    "examples": [
      "unknown, 5 per cluster",
      "unknown, 2 to 4 per cluster",
      "unknown, multiple per cluster"
    ]
  },
  {
    "form": "seizure free for a duration",
    "description": "No seizures for a stated length of
time.",
    "examples": [
      "seizure free for 6 month",
      "seizure free for 2 year",
      "seizure free for 3 months",
      "seizure free for 2 years",
      "seizure free for 2 to 3 months"
    ]
  },
  {
    "form": "seizure free for a vague duration",
    "description": "No seizures for an unstated length
of time.",
    "examples": [
      "seizure free for multiple month",
      "seizure free for multiple year"
    ]
  }
]

```

```

    ],
    {
      "form": "unknown",
      "description": "Seizures are discussed, but there is
no usable frequency.",
      "examples": [
        "unknown"
      ]
    },
    {
      "form": "no seizure frequency reference",
      "description": "There is no usable frequency
evidence.",
      "examples": [
        "no seizure frequency reference"
      ]
    }
  ],
  "event_schema": {
    "event_id": "stable string such as e1",
    "kind": [
      "frequency_rate",
      "cluster_frequency",
      "seizure_free",
      "last_event_only",
      "unknown_frequency",
      "no_reference"
    ],
    "raw_value": "source-near expression or null",
    "applies_to": "seizure type or clinical target, or null",
    "time_window": "source-near current/recent/historical
window, or null",
    "temporality": [
      "current",
      "recent",
      "historical",
      "future",
      "unclear"
    ],
    "assertion_status": [
      "asserted",
      "negated",
      "historical",
      "hypothetical",
      "unknown"
    ],
    "evidence": "exact note substring",
    "notes": "optional short note or null"
  },
  "selection_schema": {
    "selected_event_ids": "list of selected event_id strings",
    "final_kind": [
      "frequency",
      "seizure_free",
      "unknown",
      "no_reference",
      "unresolved_multiple"
    ],
    "final_label": "normalized label, or null if not
directly countable",
    "evidence": "exact note substring supporting the final
selection",
    "confidence": [
      "low",
      "medium",
      "high"
    ],
    "rationale": "brief clinical reason for selecting these
events"
  }
}

```

REFERENCES

- [1] S. T. Rosenbloom, J. C. Denny, H. Xu, N. Lorenzi, W. W. Stead, and K. B. Johnson, "Data from clinical notes: A perspective on the tension between structure and flexible documentation," *J. Amer. Med. Informat. Assoc.*, vol. 18, no. 2, pp. 181–186, 2011.

- [2] D. A. Hanauer, Y. Liu, Q. Mei, F. J. Manion, U. J. Balis, and K. Zheng, "Hedging their Mets: The use of uncertainty terms in clinical documents and its potential implications when sharing the documents with patients," in *AMIA Annu. Symp. Proc.*, 2012, pp. 321–330.
- [3] B. Fonferko-Shadrach *et al.*, "Annotation of epilepsy clinic letters for natural language processing," *J. Biomed. Semantics*, vol. 15, no. 17, 2024.
- [4] B. Fonferko-Shadrach *et al.*, "Using natural language processing to extract structured epilepsy data from unstructured clinic letters: development and validation of the ExECT (extraction of epilepsy clinical text) system," *BMJ Open*, vol. 9, no. 4, 2019, Art. no. e023232.
- [5] Y. Gan *et al.*, "Reproducible synthetic clinical letters for seizure frequency information extraction," arXiv:2603.11407, 2026.
- [6] K. Xie *et al.*, "Extracting seizure frequency from epilepsy clinic notes: a machine reading approach to natural language processing," *J. Amer. Med. Informat. Assoc.*, vol. 29, no. 5, pp. 873–881, 2022.
- [7] B. Holgate, S. Fang, A. Shek, M. McWilliam, P. Viana, J. S. Winston, J. T. Teo, and M. P. Richardson, "Extracting epilepsy patient data with Llama 2," in *Proc. 23rd Workshop on Biomedical Natural Language Processing (BioNLP)*, 2024, pp. 526–535.
- [8] B. Holgate, J. Davies, S. Fang, J. S. Winston, J. T. Teo, and M. P. Richardson, "Fine-tuning LLMs to extract epilepsy seizure frequency data from health records," in *Proc. 24th Workshop on Biomedical Language Processing (BioNLP)*, 2025, pp. 44–55.
- [9] B. M. Decker *et al.*, "Development of a natural language processing algorithm to extract seizure types and frequencies from the electronic health record," *Seizure*, vol. 101, pp. 48–51, 2022.
- [10] M. Fernandes, A. Cardall, L. M. V. R. Moura, C. McGraw, S. F. Zafar, and M. B. Westover, "Extracting seizure control metrics from clinic notes of patients with epilepsy: A natural language processing approach," *Epilepsy Res.*, vol. 207, 2024, Art. no. 107451.
- [11] T. Brown *et al.*, "Language models are few-shot learners," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 33, 2020, pp. 1877–1901.
- [12] M. Agrawal, S. Hegselmann, H. Lang, Y. Kim, and D. Sontag, "Large language models are few-shot clinical information extractors," in *Proc. Conf. Empirical Methods in Natural Language Processing (EMNLP)*, 2022, pp. 1998–2022.
- [13] R. Grishman and B. Sundheim, "Message Understanding Conference-6: A brief history," in *Proc. 16th Int. Conf. Computational Linguistics (COLING)*, 1996, pp. 466–471.
- [14] Z. Yang *et al.*, "CLINES: Clinical LLM-based information extraction and structuring agent," medRxiv:10.64898/2025.12.01.25341355, 2025.
- [15] S. Fang, B. Holgate, A. Shek, J. S. Winston, M. McWilliam, P. F. Viana, J. T. Teo, and M. P. Richardson, "Extracting epilepsy-related information from unstructured clinic letters using large language models," *Epilepsia*, vol. 66, pp. 3369–3384, 2025.
- [16] S. Wigglesworth, A. Neligan, J. M. Dickson, A. Pullen, E. Yelland, T. Anjuman, and M. Reuber, "The incidence and prevalence of epilepsy in the United Kingdom 2013–2018: A retrospective cohort study of UK primary care data," *Seizure*, vol. 105, pp. 37–42, 2023.
- [17] R. S. Fisher *et al.*, "ILAE official report: A practical clinical definition of epilepsy," *Epilepsia*, vol. 55, no. 4, pp. 475–482, 2014.
- [18] K. Xie, B. Litt, D. Roth, and C. A. Ellis, "Quantifying clinical outcome measures in patients with epilepsy using the electronic health record," in *Proc. BioNLP Workshop*, 2022, pp. 369–375.
- [19] L. Cui, S. S. Sahoo, S. D. Lhatoo, G. Garg, P. Rai, A. Bozorgi, and G.-Q. Zhang, "Complex epilepsy phenotype extraction from narrative clinical discharge summaries," *J. Biomed. Informat.*, vol. 51, pp. 272–279, 2014.
- [20] S. Fu *et al.*, "Clinical concept extraction: A methodology review," *J. Biomed. Informat.*, vol. 109, 2020, Art. no. 103526.
- [21] Y. Wang *et al.*, "Clinical information extraction applications: A literature review," *J. Biomed. Informat.*, vol. 77, pp. 34–49, 2018.
- [22] L. Chiticariu, Y. Li, and F. Reiss, "Rule-based information extraction is dead! Long live rule-based information extraction systems!," in *Proc. Conf. Empirical Methods in Natural Language Processing (EMNLP)*, 2013, pp. 827–832.
- [23] R. E. Sherman *et al.*, "Real-world evidence—what is it and what can it tell us?" *N. Engl. J. Med.*, vol. 375, no. 23, pp. 2293–2297, 2016.
- [24] J. Wang *et al.*, "Scaling clinician-grade feature generation from clinical notes with multi-agent language models," arXiv:2508.01956, 2025.
- [25] A. Gorenshstein *et al.*, AI Agents in Clinical Medicine: A Systematic Review
- [26] D. Tran, Single-Agent LLMs Outperform Multi-Agent Systems on Multi-Hop Reasoning Under Equal Thinking Token Budgets